

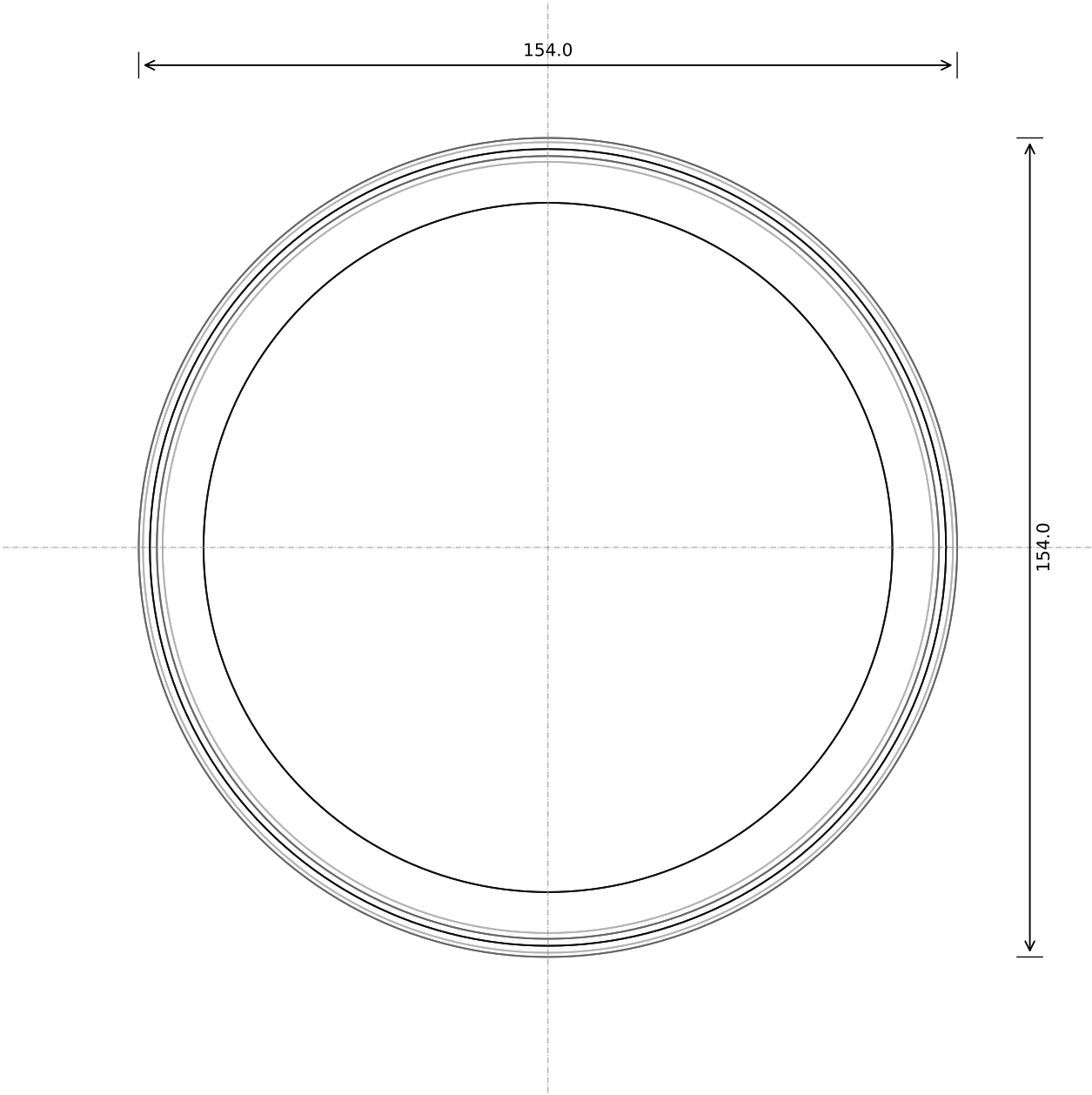
the 60 — v10 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-05. Stack: plate 8.0, deck 19.48-20.98, seat 30.4, glass 30.90-32.88, lens to 35.38, knob top 38.78, overall 40.28 with the pad. Ø154.

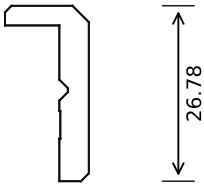
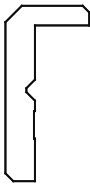
1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board: 65.0 x 64.0 x 1.6, parts 6.5	DM-ADTTR-014 outline PUBLISHED; socket position and height assumed
10. audio / motion / touch / converter boards: envelopes	electrical layout not done
11. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
12. right-angle barrel plug: Ø8.5 body	measure the brick's plug
13. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
14. MOTOR_BAND_T: 0.6	carried from v9
15. servo pushrod: 4 mm bridge to the tab	carried from v9; servo now on the deck

Build-time changes against V10-SPECIFICATION.md are listed in V10.md. Sheets 1-11 follow: one per made part.

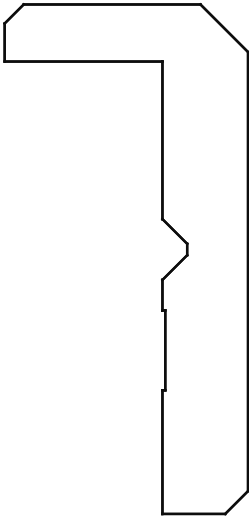
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 47 to 79, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- outside Ø154.0, bore Ø145.0 (0.45 to the panel's chin corners)
- lip inner edge Ø128.4, lip 3.0 thick, top at z 39.18
- outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
- V-groove 90°, root r 73.8, z 24.3-27.5
- code-band recess 0.15 deep, z 18.5-22.7
- diamond knurl 68 starts x 2, 1.0 deep, helix 33.5°, 4 whole rows z 14.0-35.5
- wall under the knurl root 3.5

the 60 — v10

Knob

Cadrane · sheet 1 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)

Process: print top face down, 0.16 mm; no supports

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

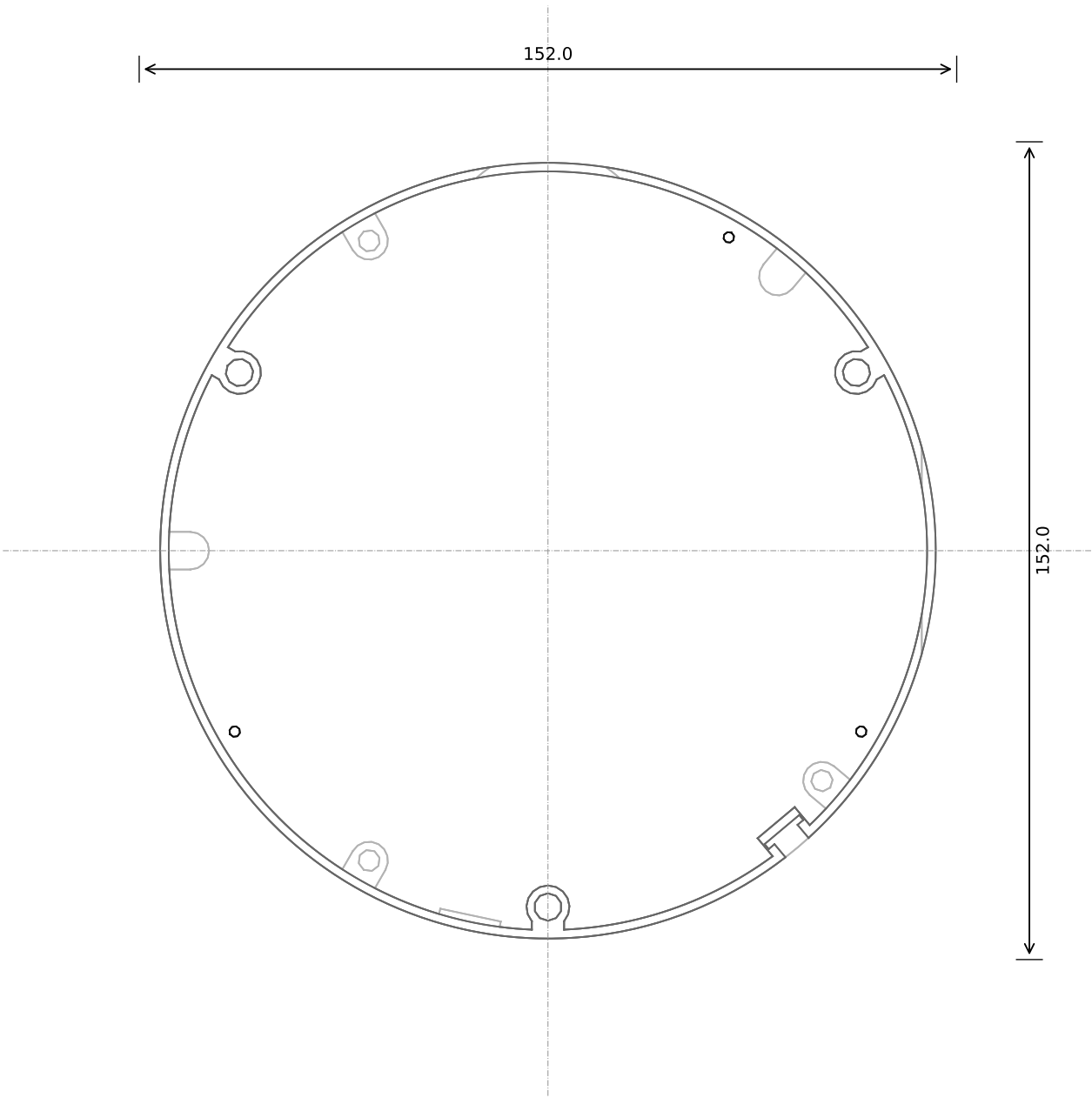
Part: knob_body

Overall 154.0 x 154.0 x 26.8

Volume 63.82 cm³

Scale: see each view (fit)

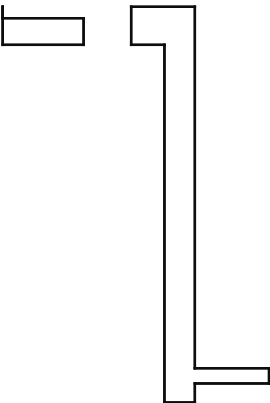
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 46 to 78, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

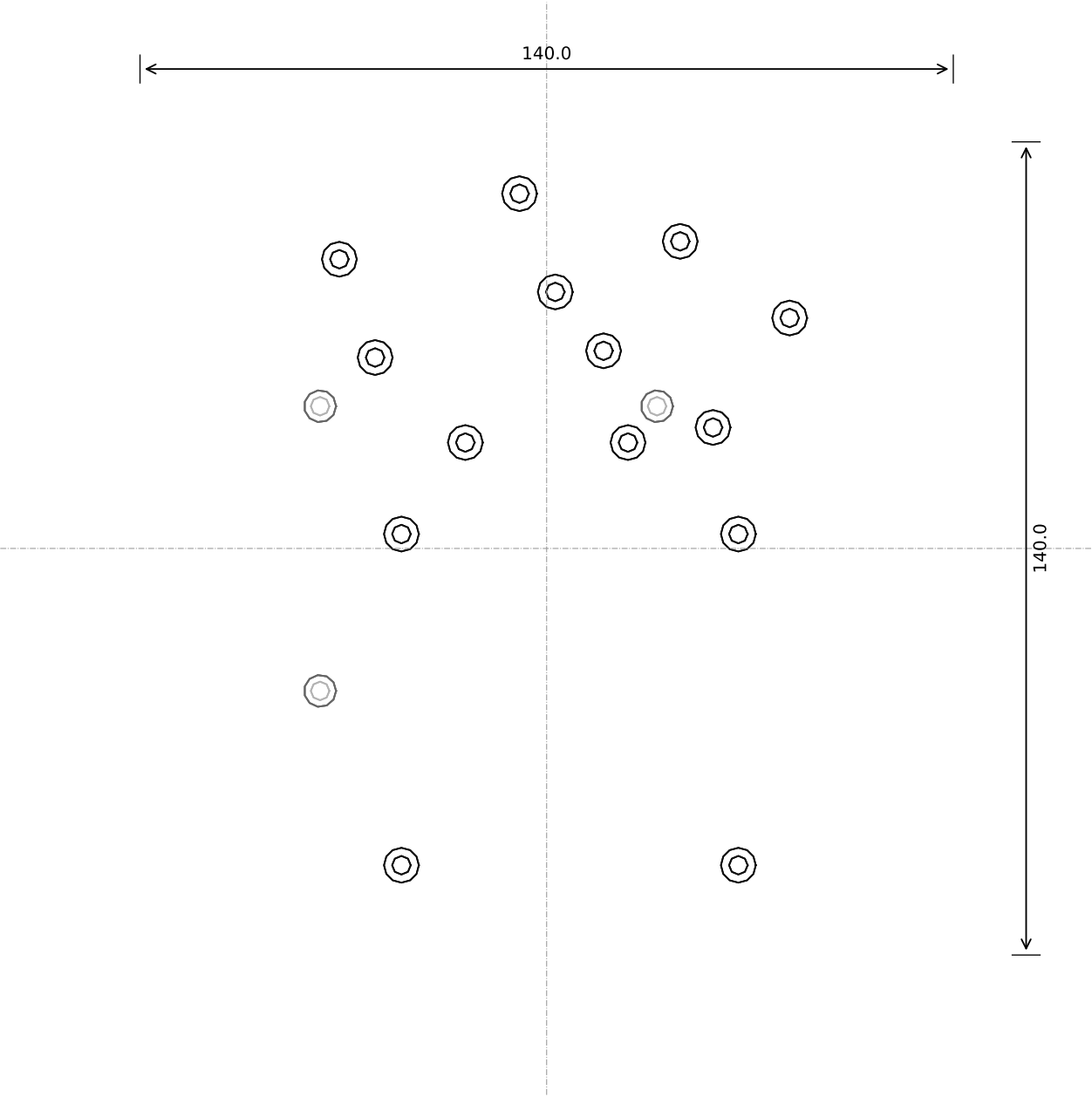
- 1. wall r 70.5-72.1, z 8.0-30.4
- 2. seat flange r 62.0-72.1, z 28.4-30.4; three Ø2 locating nubs at [60, 210, 330]°
- 3. diffuser lip r 72.1-76.0, z 10.6-11.4
- 4. wheel posts Ø8.0 at r 66.23, az [30, 150, 270], to z 23.3; bush bores Ø5.1; wall windows 15.2 x 5.4
- 5. pillars Ø7 at r 66.5, az [50, 120, 180, 240, 320], to z 19.48; M3 inserts top (all) and bottom ([120, 240, 320])
- 6. encoder tower at 310.0°, window through the wall at the code band; LRA pad at 258.0°
- 7. flex slot 32 x 2.5 through the seat at 0°, offset -4; 16 ports Ø4.0 at z 15.0
- 8. motor relief at 90°

the 60 — v10 Internal structure

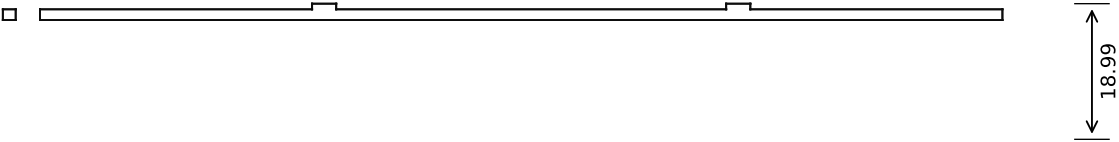
Cadrane · sheet 2 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / PA12 (production)	Part: internal_structure
Process: print upright on the wall foot, 0.20 mm; support the seat flange under the 152.0 x 152.0 windows	Overall 152.0 x 152.0 x 24.6
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,	Volume 27.52 cm³
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

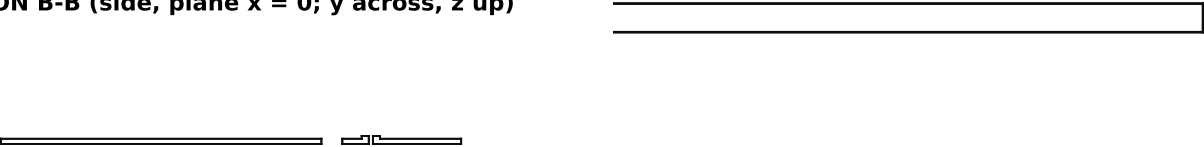


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 40 to 72, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. disc Ø140 x 1.5, z 19.48-20.98
- 2. three legs Ø5.5 down to the Pi's board top (z 2.7800000000000002), M2.5 inserts
- 3. tab slot 6.0 x 10.6 at 90°; ribbon slot 22 x 3 at (-16.5, -34.5)
- 4. Ø3.4 holes at the pillars; notches at the wheel posts and the encoder tower
- 5. M2.5 insert bosses for the adapter, audio, motion boards and the servo tray

the 60 — v10

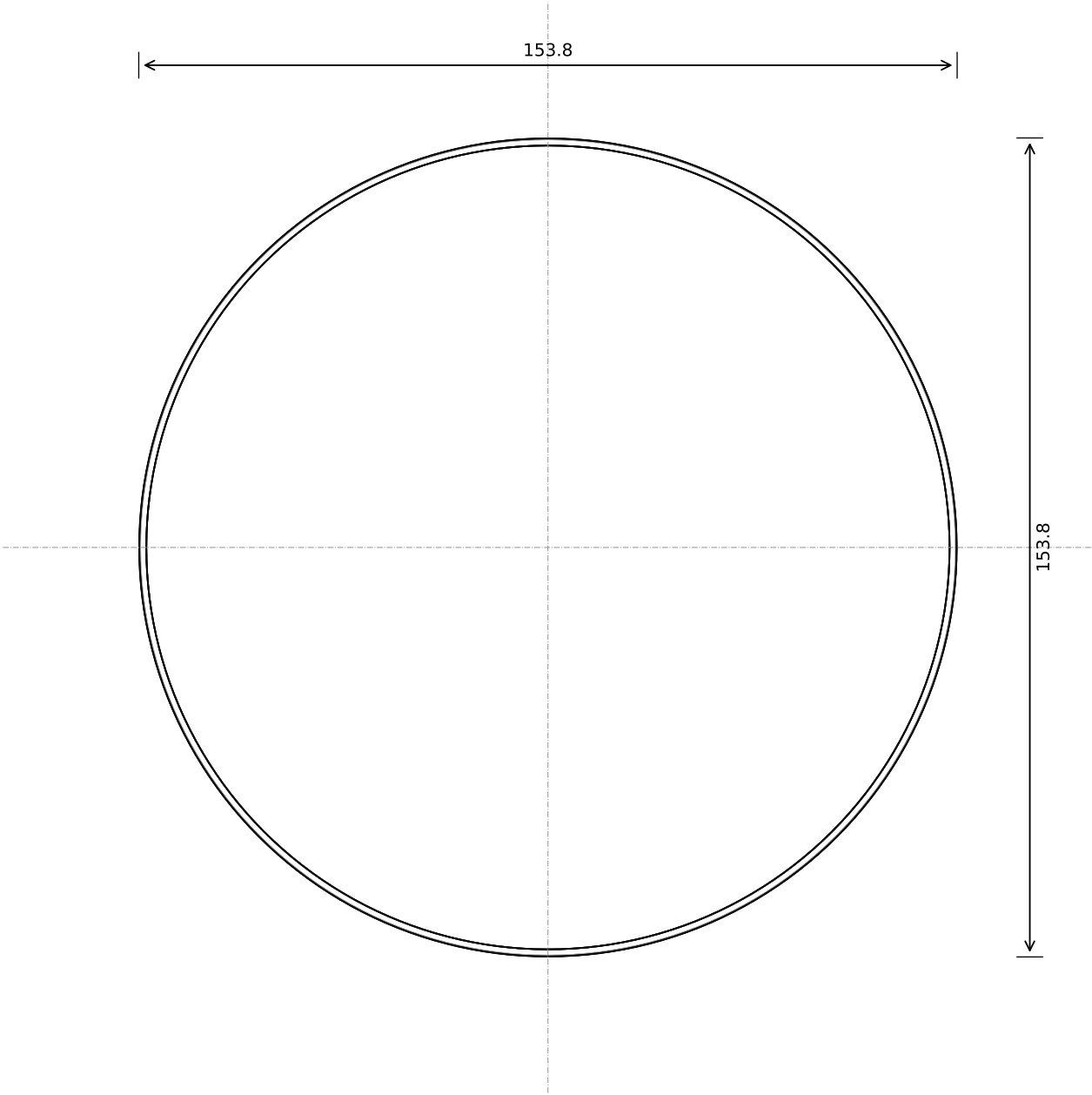
Mezzanine deck

Cadrane · sheet 3 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: print upside down (legs up), 0.20 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: deck
Overall 140.0 x 140.0 x 19.0
Volume 23.63 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 47 to 79, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. r 75.5-76.6 at the bottom, to r 77.0 at the top; z 8.0-10.6
- 2. 0.5 chamfer on the outside top edge
- 3. retained by the structure's lip; sits on the plate's rim outside the LED flex

the 60 — v10Halo diffuser

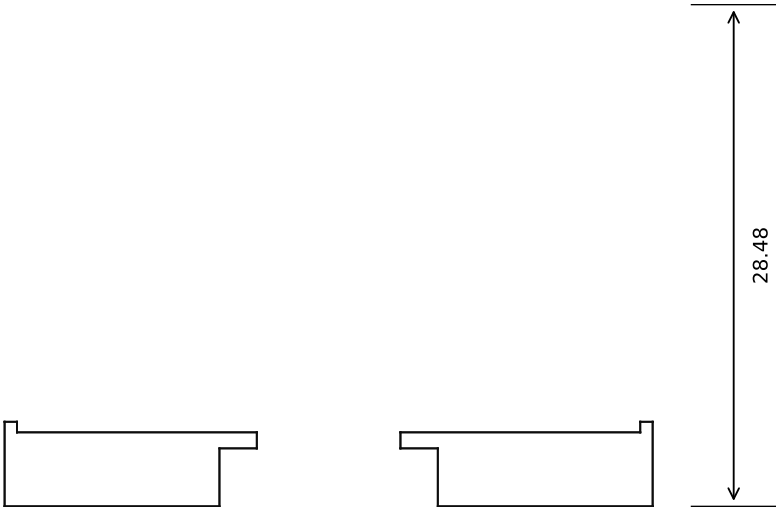
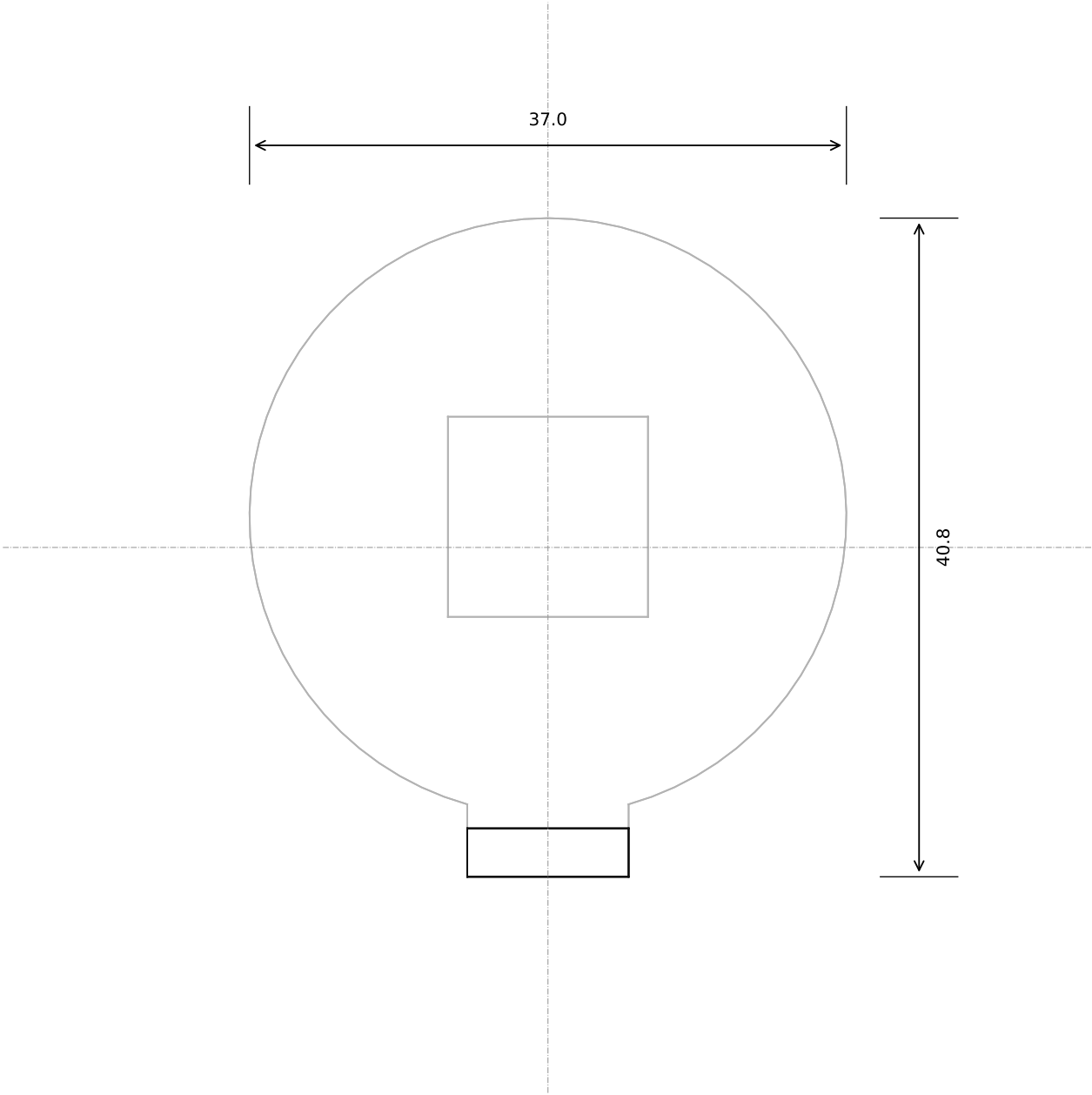
Cadrane · sheet 4 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: opal PMMA (production) / natural PETG (prototype)
Process: print on its underside, 0.20 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: halo_diffuser
Overall 153.8 x 153.8 x 2.6
Volume 1.56 cm³
Scale: see each view (fit)

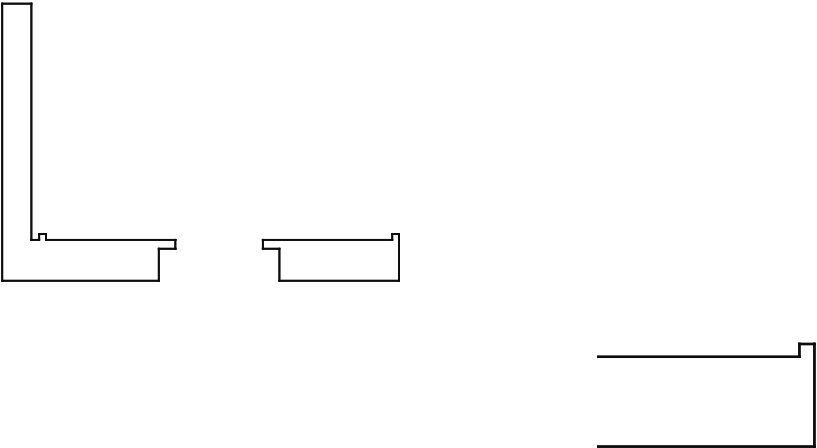
SECTION A-A (front, plane y = 0; x across, z up)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. shoe $\varnothing 37 \times 4.2$; locating rim $\varnothing 35.6/37 \times 0.6$
- 2. commutation-board pocket $12.4 \times 12.4 \times 3.3$; sight hole $\varnothing 9$
- 3. push tab 3.0×10.0 to z 28.48 (through the deck to the servo)
- 4. travel 2.4 radial in the plate's stadium hole

the 60 — v10

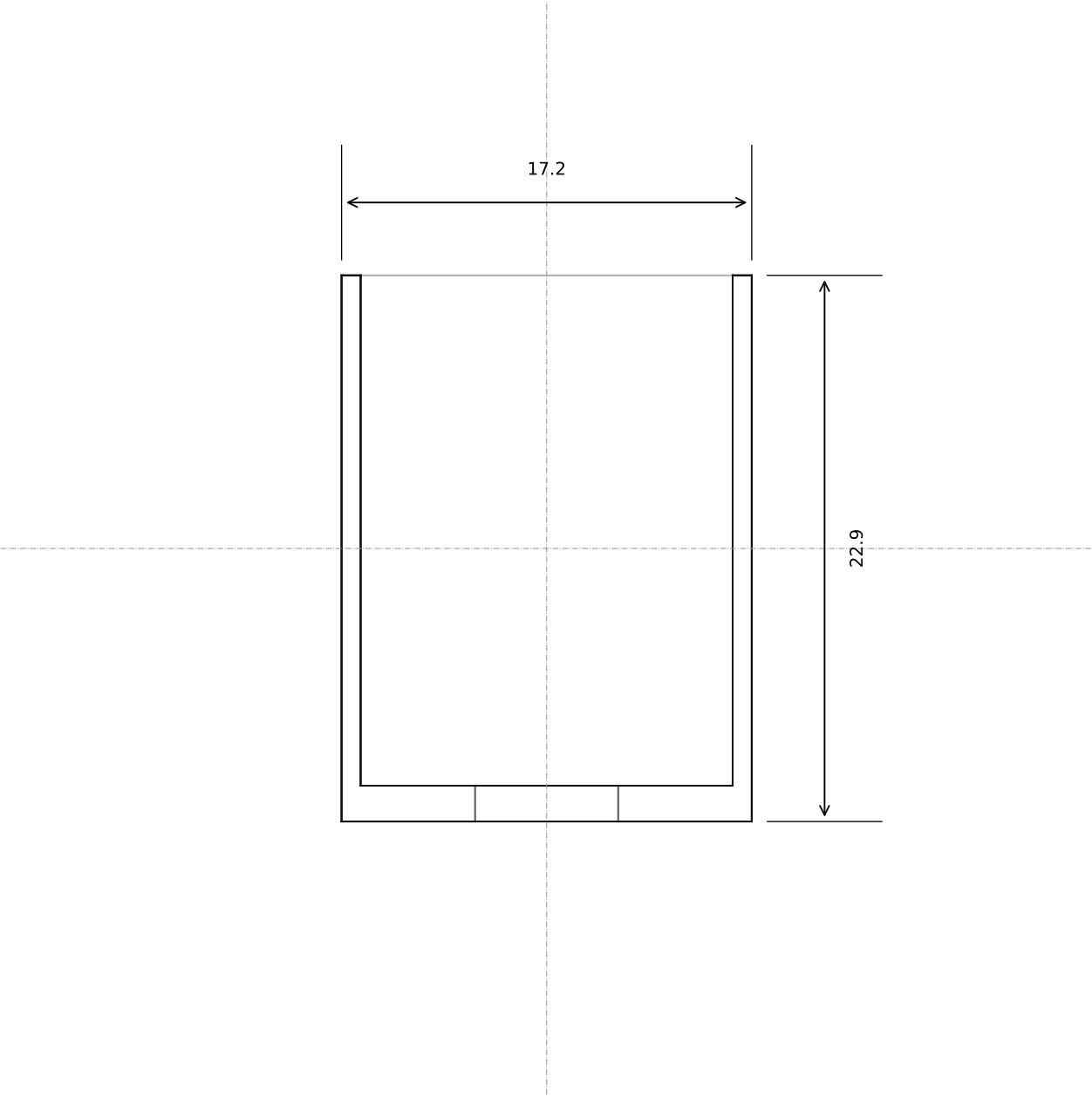
Motor carriage

Cadrane · sheet 5 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: print standing on the shoe, 0.16 mm; brace the 28 mm tab
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: carriage
Overall $37.0 \times 40.8 \times 28.5$
Volume 4.90 cm³
Scale: see each view (fit)

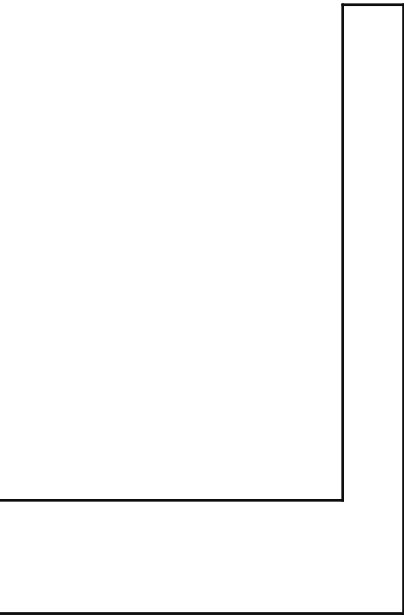
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



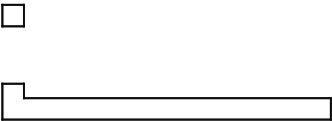
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 4 to 11, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. tray for the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, floor 1.5
- 2. open toward the carriage tab; lead exit at the inner end
- 3. 2 x M2.5 into deck inserts

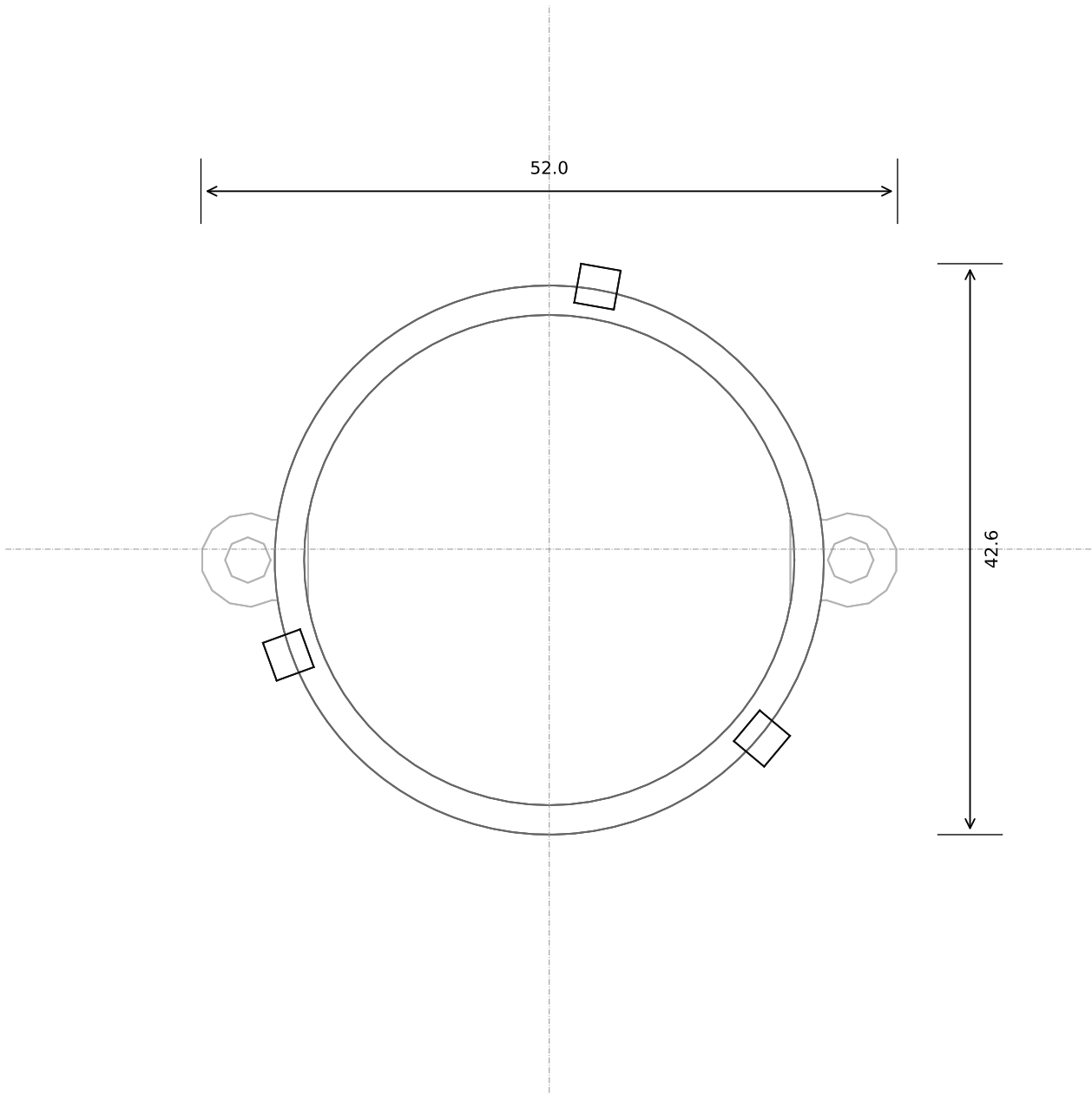
the 60 — v10 Servo tray (on the deck)

Cadrane · sheet 6 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: print flat, 0.20 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: servo_mount
Overall 17.2 x 22.9 x 8.0
Volume 0.94 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



Key dimensions (from the parameter block):

- 1. ring r 18.3-20.5 x 4.45 under the Ø40 flange
- 2. three snap fingers at 80/200/320° (local)
- 3. two ears at ±22.5 on the local x axis, M3 countersunk from below the plate

SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 13 to 28, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



the 60 — v10

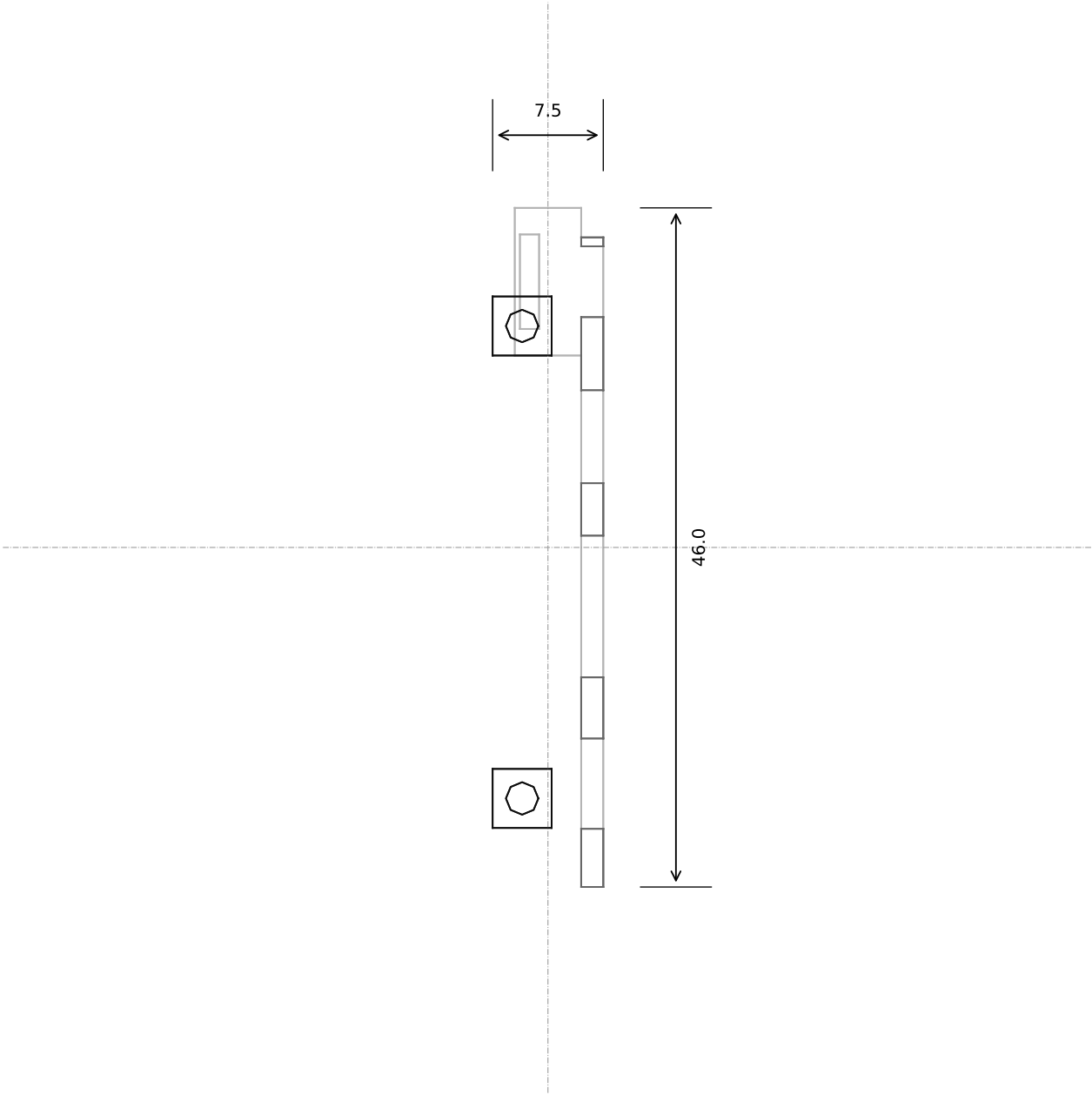
Speaker cradle

Cadrane · sheet 7 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

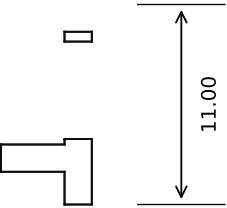
Material: PETG
Process: print flat, 0.20 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: speaker_cradle
Overall 52.0 x 42.6 x 7.2
Volume 1.36 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



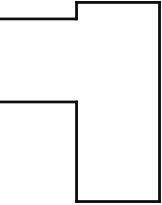
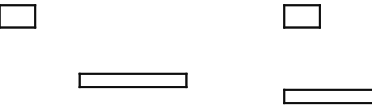
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 2 to 6, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 44.0 wide x 1.5 thick at r 69.0, z -1.5-8.0
- 2. barrel nose Ø6.2 at t -15.0, z 3.5; USB-C opening at t -3.0, z 4.5; jack Ø6.3 at t 8.5, z 4.0; light aperture Ø4.8 at t 19.0
- 3. shelf for the USB-C board; foot for the light-sensor board; two ears at t ±16 on the plate top (M2)

the 60 — v10

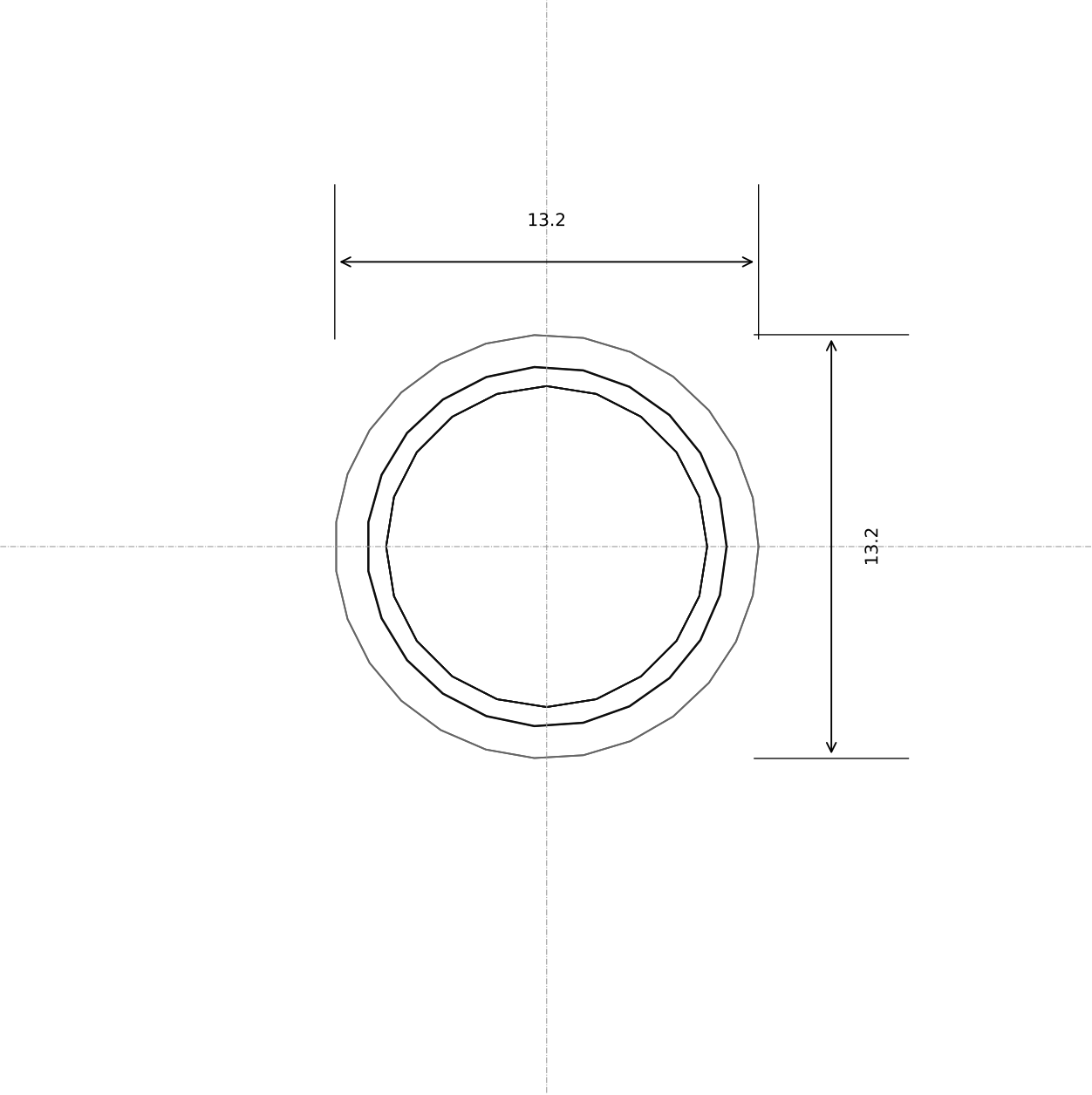
Rear port face

Cadrane · sheet 8 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: print standing, 0.16 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: port_face
Overall 7.5 x 46.0 x 11.0
Volume 0.63 cm³
Scale: see each view (fit)

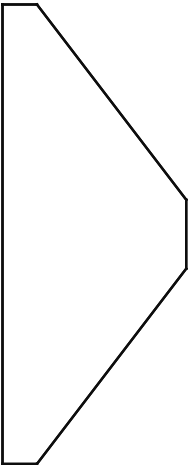
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø13.2 x 4.0, 90° V 1.3 deep with a 0.6 flat; bore Ø10 press over a 623ZZ

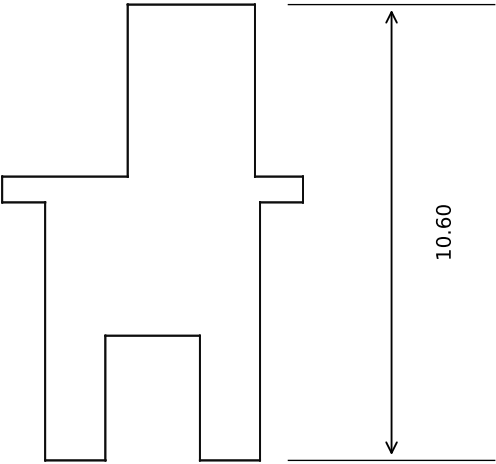
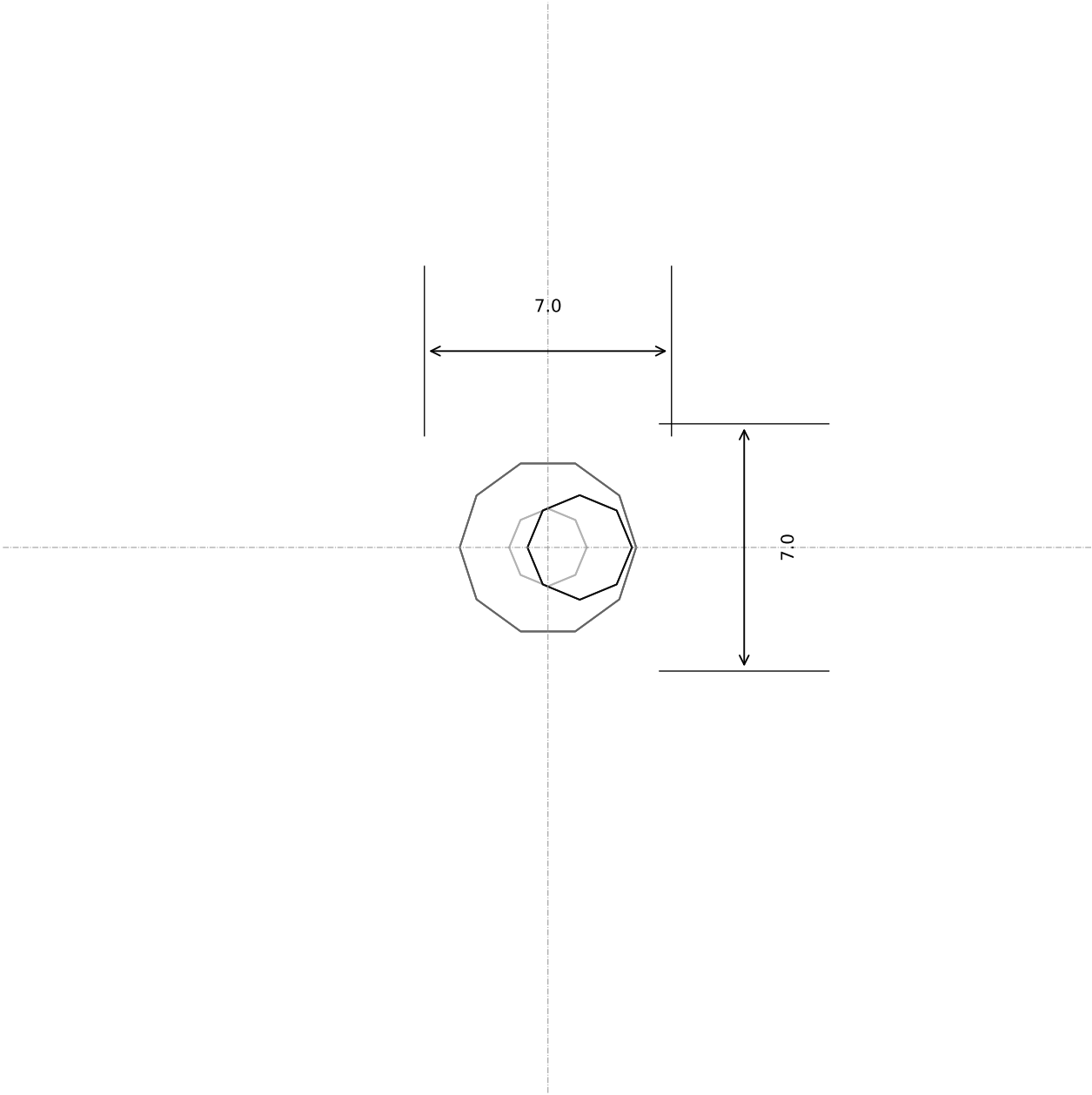
the 60 — v10 Wheel V-collar (x3)

Cadrane · sheet 9 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / POM (production)
Process: print axis vertical, 0.12 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

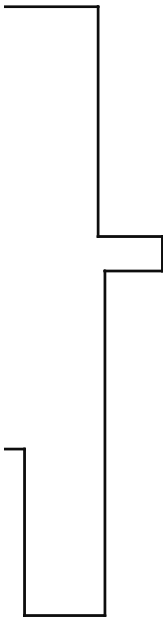
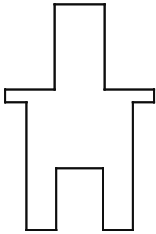
Part: collar_0
Overall 13.2 x 13.2 x 4.0
Volume 0.15 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



DETAIL C (section A-A, x 2 to 6, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø5.0 shank x 6, Ø7.0 x 0.6 flange, Ø3.0 pin x 4.0 offset 0.9; 2 mm hex socket

the 60 — v10

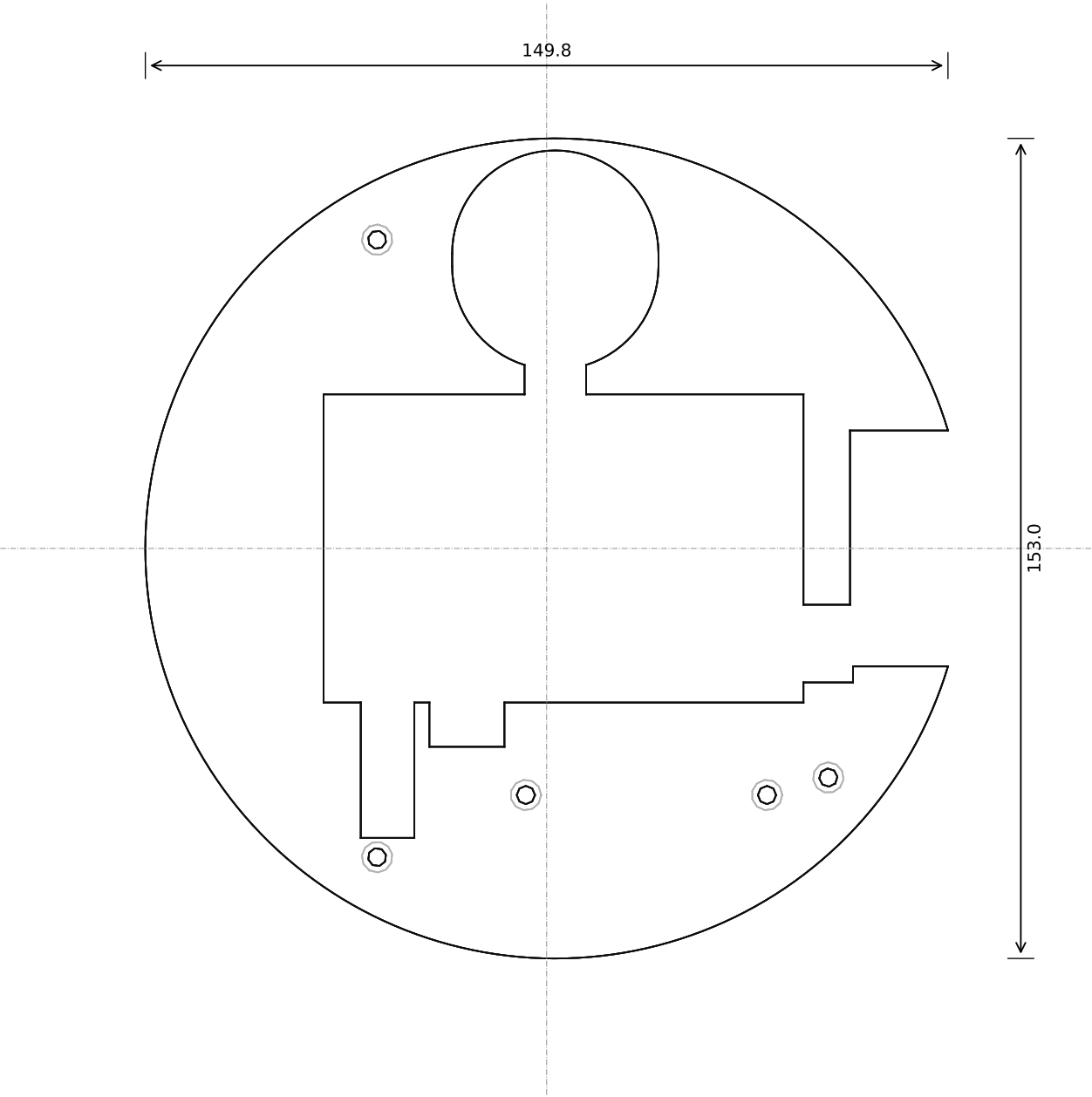
Eccentric bush (x3)

Cadrane · sheet 10 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / brass (production)
Process: print pin up, 0.12 mm
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: bush_0
Overall 7.0 x 7.0 x 10.6
Volume 0.16 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 45 to 77, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. Ø153 x 8.0
- 2. Pi window 89.5 x 57.5 with plug notches; carriage stadium hole; port slot 44.0 wide from r 55.0
- 3. 3 x M3 countersunk from below at r 66.5, az [120, 240, 320]; 2 x M3 csk for the speaker cradle; ground-bond hole; 2 x Ø1.6 for the port-face ears

the 60 — v10

Base plate (steel)

Cadrane · sheet 11 of 11 · 2026-09-05 · drawn from src/params.py + model.py (build123d)

Material: mild steel 8 mm, laser/waterjet cut, powder-coated
Process: not printed (print a stand-in if the steel is not back)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: base_plate
Overall 149.8 x 153.0 x 8.0
Volume 83.78 cm³
Scale: see each view (fit)